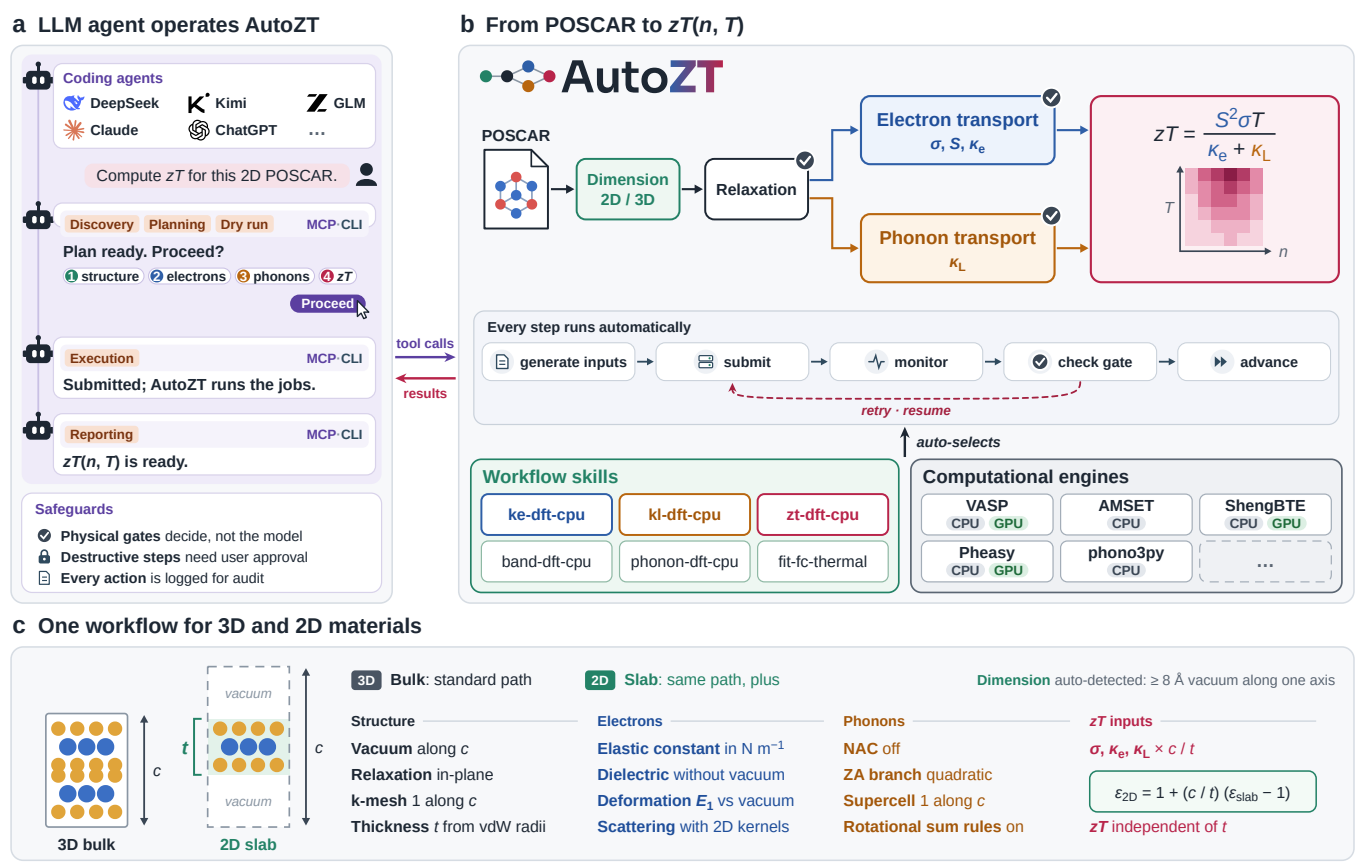

a LLM agent operates AutoZT
Coding agents
DeepSeek
Kimi
GLM
Claude
ChatGPT
...
Compute zT for this 2D POSCAR.
Discovery
Planning
Dry run
MCP·CLI
Plan ready. Proceed?
1 structure
2 electrons
3 phonons
4 zT
Proceed
Execution
MCP·CLI
Submitted; AutoZT runs the jobs.
Reporting
MCP·CLI
zT(n, T) is ready.
Safeguards
Physical gates decide, not the model
Destructive steps need user approval
Every action is logged for audit
tool calls
results
b From POSCAR to zT(n, T)
AutoZT
POSCAR
Dimension
2D / 3D
Relaxation
Electron transport
σ, S, κe
Phonon transport
κL
zT = S²σT / (κe + κL)
T
n
Every step runs automatically
generate inputs
submit
monitor
check gate
advance
retry · resume
auto-selects
Workflow skills
ke-dft-cpu
kl-dft-cpu
zt-dft-cpu
band-dft-cpu
phonon-dft-cpu
fit-fc-thermal
Computational engines
VASP
CPU GPU
AMSET
CPU
ShengBTE
CPU GPU
Pheasy
CPU GPU
phono3py
CPU
...
c One workflow for 3D and 2D materials
vacuum
vacuum
c
t
c
3D bulk
2D slab
3D Bulk: standard path
2D Slab: same path, plus
Dimension auto-detected: ≥ 8 Å vacuum along one axis
Structure
Vacuum along c
Relaxation in-plane
k-mesh 1 along c
Thickness t from vdW radii
Electrons
Elastic constant in N m⁻¹
Dielectric without vacuum
Deformation E1 vs vacuum
Scattering with 2D kernels
Phonons
NAC off
ZA branch quadratic
Supercell 1 along c
Rotational sum rules on
zT inputs
σ, κe, κL × c / t
ε2D = 1 + (c / t) (εslab − 1)
zT independent of t